

Anomaly detection of a Ship's Engine

Methods, Insights and Operational implications

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Problem Statement

Modern ship engines use sensors to collect large amounts of data across multiple parameters including RPM, temperature and pressure readings. Traditional monitoring methods are mostly reactive as they use preventive maintenance procedures, thereby limiting the ability to identify unusual behaviour, leading to increased maintenance costs, unplanned downtime and reduced engine efficiency.

This report addresses this need using various models that uncover hidden patterns thereby providing insights for predictive maintenance and decision-making

Introduction

Dataset used: 19,535 observations(rows) and 6 features(columns)

Purpose: Anomaly detection

Assumption: Anomalies would make up 1-5% of the data

Methods:

Method	Type	Technique
IQR	Statistical	Flags values outside the Inter Quartile Range (typically +/- 1.5xIQR)
One-Class SVM	Unsupervised(Kernel based)	Flags values outside a boundary that it makes by learning about normal data
Isolation Forest	Unsupervised(Tree-based)	Flags values by randomly partitioning data using splits

Table 1: Anomaly detection methods

Methodology

1. Approach

The overall approach involved:

- Data assessment and Exploration (EDA)
- Anomaly detection using IQR
- Anomaly detection using One-class SVM and Isolation Forest
- Comparison of the models

2. Workflow

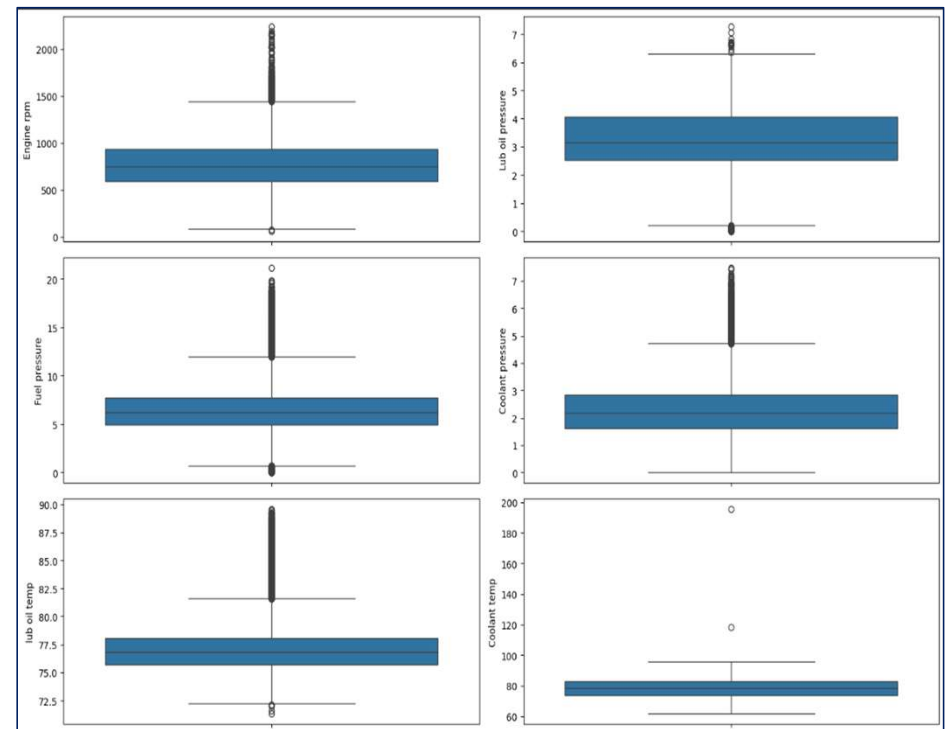
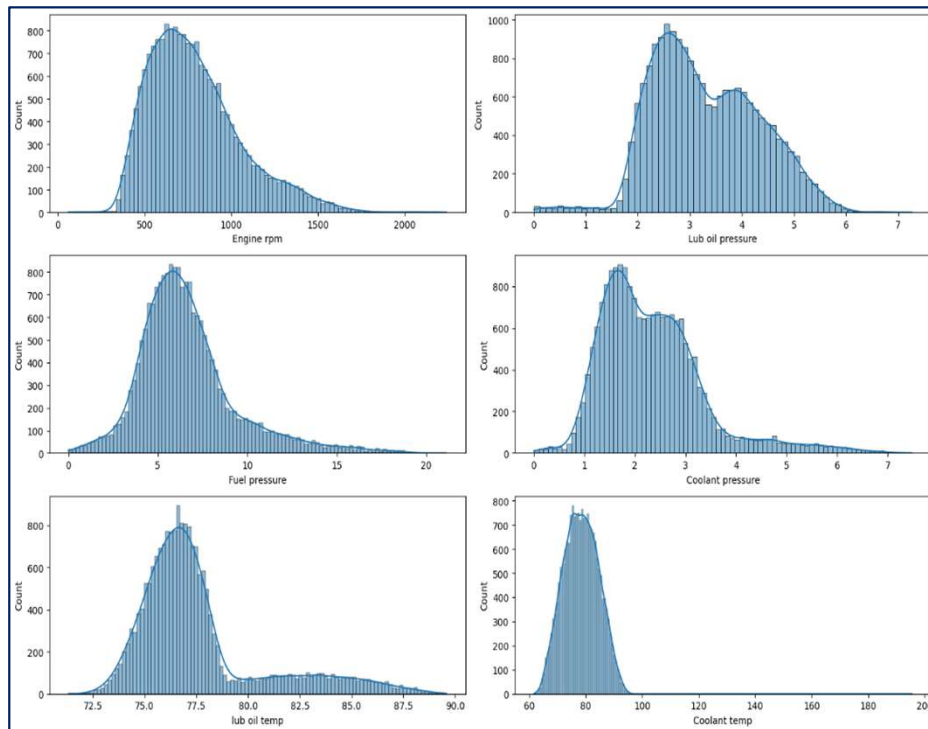
Steps	Description
Data assessment & Exploration	• Checking: Size and Structure of data • Identifying: Missing and Duplicate values • Evaluating: Descriptive statistics • Creating: Visuals using Histograms and Boxplots
Anomaly detection using IQR	• Anomalies were identified for each row • Features were identified that had the highest number of anomalies • Different combination of number of features were evaluated to assess the percentages of observations classified as outliers (see Table 2)
Anomaly detection using One-Class SVM	• Multiple combinations of Gamma and Nu inputs were evaluated to get the optimum values for the model (see Table 3) • Scaling was performed using Standardisation. • Dimensionality reduction was performed using PCA for visualising the output in the 2D space.
Anomaly detection using Isolation Forest	• Contamination factor directly controls the proportion of anomalies. • Dimensionality reduction was performed using PCA for visualising the output in the 2D space.

Results

Interpretation of plots

Most readings fall within a normal operating range but occasional spikes occur which may reflect abnormal operation, sensor data issues etc.

High number of anomalies are found above the upper bound limit and exist specifically in Engine RPM, Fuel pressure, Coolant pressure and Lubricating oil temperatures.



Evaluation of Methods

IQR Method:

Evaluation of the number of features and the % of anomalies produced the results as shown in Table 2.

23.7% anomalies were observed in the observations with at least **one** feature.

2.16% anomalies were observed in the observations with at least **two** feature.

0.06% anomalies were observed in the observations with at least **two** feature.

No observations had anomalies when **≥ four** features were considered.

Based on our expected range of 1-5%, we classified observations as anomalies if they had outliers in **≥ two** features. Total **422** anomalies were detected.

Lubricating oil temperature and Fuel pressure were identified as the features with the highest number of anomalies.

One-Class SVM:

Different ranges of anomalies were found based on combinations of parameters, as shown in Table 3.

Low Gamma at Low Nu – Detects fewer anomalies (e.g., 0.97%) ➡ **Makes the model less sensitive to outliers**

High Gamma at Low Nu – Detects more anomalies (e.g., 3.13%) ➡ **Makes the model more sensitive to outliers**

Also, As Nu increases, the model flags more points as anomalies.

We considered **Gamma=0.1** and **Nu=0.05** for our model, detecting **978** anomalies.

Isolation Forest:

For Isolation Forest model, contamination factor of **0.05** was considered for our model, detecting **977** anomalies

Table 2: Anomaly detection methods

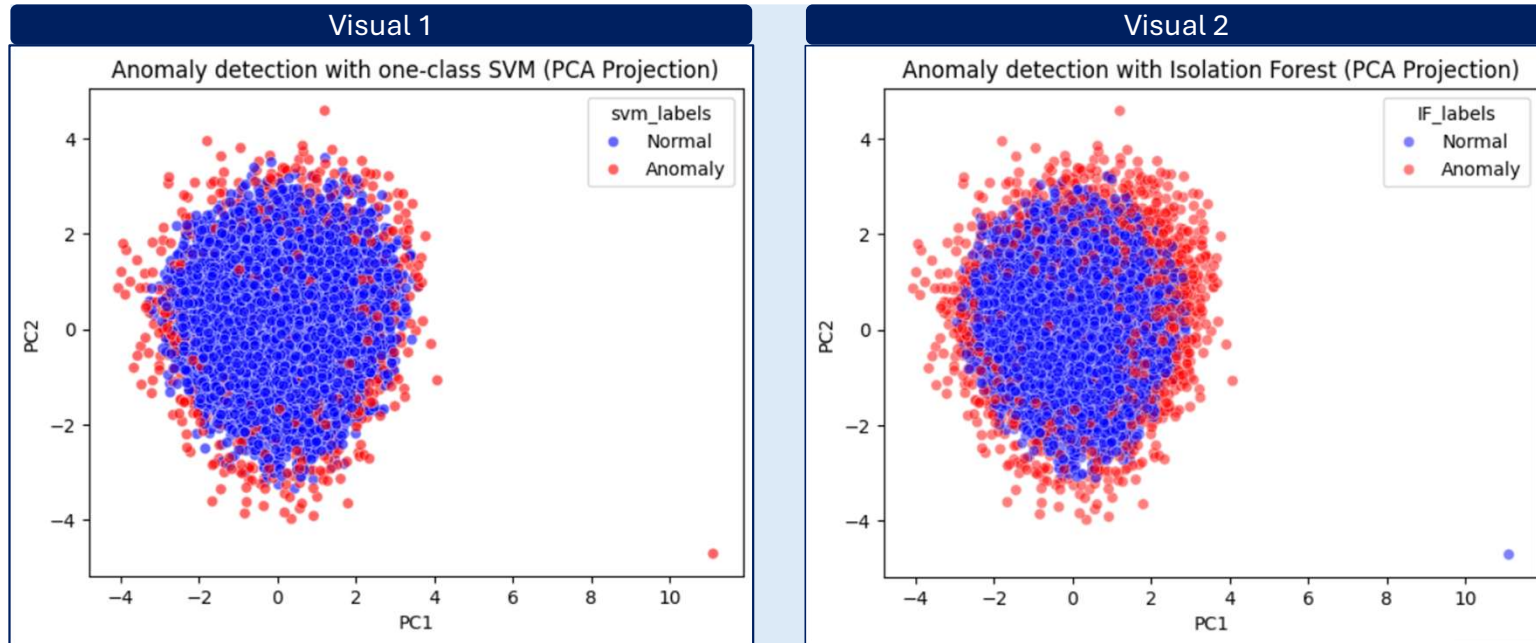
Number of features	% of anomalies
≥ 1	23.7
≥ 2	2.16
≥ 3	0.06
≥ 4	0.00

Table 3: Different ranges of anomalies found based on different parameters

Gamma	Range of Nu	Range of Anomalies found
0.1	0.01 - 0.05	0.97% - 5.01%
0.2	0.01 - 0.05	1.03% - 5.03%
0.3	0.01 - 0.05	1.41% - 5.02%
0.4	0.01 - 0.05	2.22% - 5.10%
0.5	0.01 - 0.05	3.13% - 5.34%

Visualising the results

Visuals for One-Class SVM and Isolation Forest were created using the PCA method. A pattern of outliers accumulating around the normal points was observed.



Comparing results of all three methods

When results of all three methods were compared, we found around **233 (1.2%)** observations were flagged by all of them together.

Conclusion

IQR method is simple and fast but we had to keep the threshold to 2 or more features to keep the proportion of outliers within our goal.

One-Class SVM gave us 978 anomalies and visuals showed clear separation of anomalies around the cluster of normal data.

Isolation Forest gave us 977 anomalies which are nearly the same as those provided by One-Class SVM but it provided the most consistent results indicated by strong overlap with other methods, especially One-Class SVM. Other factors that makes Isolation Forest better is robust detection of anomalies (needs fewer splits) and lower sensitivity to feature scaling.

Hence, we would recommend Isolation Forest to be used for timely detection of anomalies which would help in efficient maintenance activity.

Abbreviations

Abbreviation	Full Form
IQR	Inter Quartile Range
SVM	Support Vector Machine
ML	Machine Learning
EDA	Exploratory Data Analysis
PCA	Principal Component Analysis
RPM	Revolutions per minute
2D	Two dimensional